The form of dissipation shapes what a quantum reservoir remembers

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Dissipation sets how long a quantum reservoir computer remembers its past inputs, and its rate is the usual control. Here we show that its form matters as much. When the qubits decay independently, stronger loss shortens the memory. When they decay collectively, the quantum Zeno effect holds the system in dark states that do not decay, and stronger loss prolongs the memory. A longer memory, however, leaves more time for new inputs to mix with the inputs already stored. We prove that the number of inputs mixed into a stored signal is, on average, the memory lifetime times the fraction of the signal that one input disturbs. We also prove that this number stays fixed, however long the memory lasts, when the input amplitude falls as the inverse square root of the lifetime. A weaker input leaves a fainter signal to measure, so the form of the dissipation, the input amplitude and the measurement budget have to be chosen together.

1 Introduction

Processing information over time requires a memory of what came before. A physical system can provide such a memory because each input changes its state and leaves traces that influence its later response. Reservoir computing uses these traces to perform a task: an input stream drives a dynamical system, and a weighted sum of quantities describing its state produces the output. Only these weights are trained; this mapping is called a linear readout [1, 2]. In quantum reservoir computing (QRC), the evolving system is quantum mechanical, and the readout uses quantities obtained by measuring it [3]. The challenge is to keep earlier inputs accessible while new information continues to arrive.

The environment shapes how these traces fade. A system can lose energy or information to its surroundings through processes collectively described as dissipation [4–15]. Previous work has shown that tuning the dissipation rate improves memory, in some reservoirs with the best performance at an intermediate loss rate [6, 9]. Input, interactions, and dissipation can also be optimized together [16, 17]. The rate specifies how strongly the environment acts, but the way it acts is a separate design choice. Engineering this interaction can prepare and protect selected quantum states [18–21]. For a system processing an ongoing input stream, this raises the possibility of controlling which traces of the past survive.

This distinction becomes concrete for a reservoir of interacting spins, each a two-level quantum system. When the spins decay through separate routes into the environment, they undergo independent decay. When they share a common route, they undergo collective decay. In certain joint states, the different emission paths interfere and cancel, preventing decay through that shared route. These protected states are called dark states [22–24]. Internal dynamics can still move the system into states that do decay. If this decay is sufficiently rapid, however, it suppresses those transitions, a phenomenon known as the dissipative quantum Zeno effect [25–28]. Stronger loss therefore slows the escape of information from the protected states (Fig. 1c).

A long-lived state is not yet a usable memory, because each new input changes the same state that stores the past. A signal that initially depends on one input comes to depend on combinations of several inputs, such as their product. This mixing is useful for tasks that require relationships between inputs, but it reduces the

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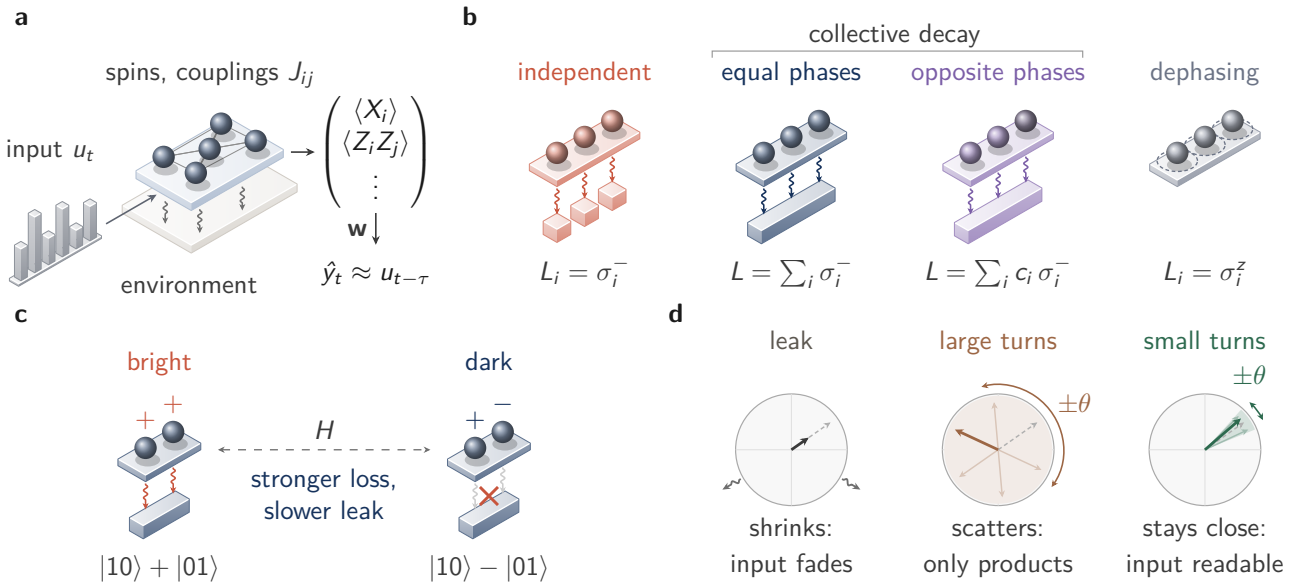


Figure 1: **Engineering how a quantum reservoir leaks. a** Inputs drive interacting spins, and a trained linear readout recalls past inputs. **b** Four forms of dissipation with the same total coupling. The darker spins couple with the opposite sign. **c** Two spins on one channel: their decays add (bright) or cancel (dark), and strong loss holds them in the dark state, the quantum Zeno effect. **d** A stored signal fades because it leaks away or because later inputs turn it at random.

capacity available for recalling single inputs [29–31]. Reading the memory has a cost as well, because resolving a weak quantum signal requires repeated preparation and measurement [10, 13, 32], and a long-lived memory is hard to read if its signal is faint. Previous work has established memory limits in classical reservoirs [2, 33, 34], explored input strength [35], and investigated how noise and protected states affect quantum information processing [36–39]. Closest to our work, Ref. [6] tunes the dissipation rate of the reservoir we study, Ref. [31] finds one trade-off between memory and nonlinearity across reservoir schemes, and Ref. [39] reads the output of a reservoir from a decoherence-free subspace to classify quantum states. The Zeno protection of dark states and the benefit of a weak input for linear memory are established. How protection, continued input, and measurement together determine the part of the past that a readout recovers is open.

Here we show that the form of the dissipation sets three properties of a memory (Fig. 1): how long a stored signal survives, whether and how strongly an input can be written into it, and how strong a signal is left to measure. First, at equal total coupling, stronger loss shortens the memory lifetime with independent decay and prolongs it with collective decay, through dark states (Sec. 3.1). Second, mixing limits how strongly an input can be written. We prove that at a fixed input amplitude a stored signal combines on average $s\,t_{\mathrm{mem}}$ inputs, where the mixing fraction s is the part of the stored signal that one input disturbs and t_{mem} is the memory lifetime (Sec. 3.2). We also prove that an amplitude that falls as $1/\sqrt{t_{\mathrm{mem}}}$ removes this limit. Single inputs then keep a fixed share of the memory, and this limit agrees with the five-spin results without a fitted parameter. The phases with which the spins couple to a collective channel decide whether an input reaches the dark states at all (Sec. 3.3). Third, the stationary state sets the size of the signal that measurements must resolve. At strong loss a collective channel whose phases differ from those of the input field recovers the most memory at every measurement budget, and where the loss can be made weak, independent decay recovers more still (Sec. 3.4).

Together, these results connect the physical survival of information to what a readout recovers from it. The protection relies on quantum interference, whereas the two theorems hold for every reservoir that meets their conditions, classical ones included, and give the size of the trade-off between memory and nonlinearity. The form of the dissipation, the input amplitude and the measurement budget become joint design parameters,

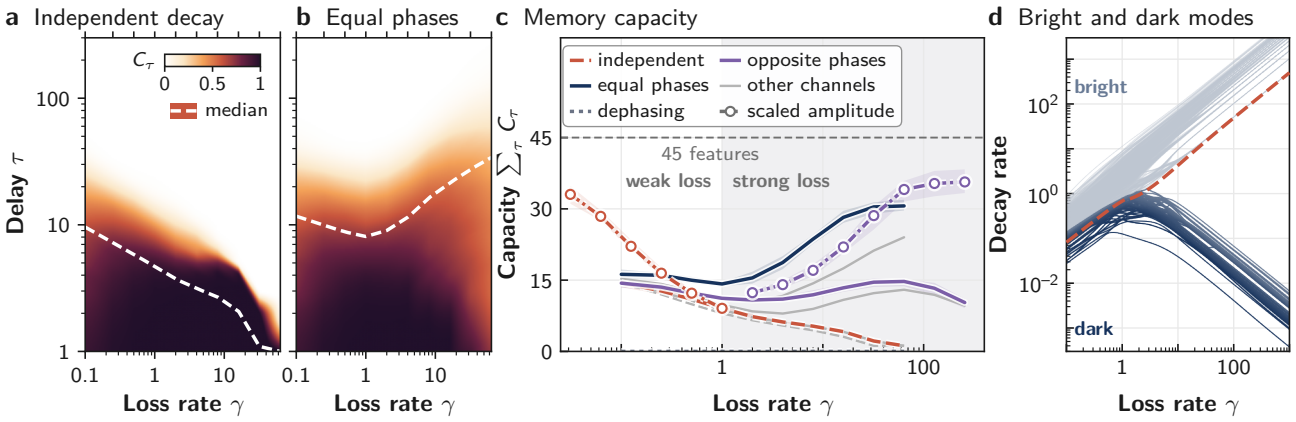


Figure 2: **The decay channel decides whether stronger loss shortens or prolongs the memory. a,b** Recall of the input τ steps back: stronger loss pushes it to shorter delays with independent decay (a) and to longer delays with the equal-phase channel (b). **c** Linear memory capacity of all channels at a fixed input amplitude, and with the scaled amplitude of Sec. 3.3 (circles), in simulated runs summed over the first 300 delays. Shading marks loss stronger than the couplings ($\gamma > 1$). **d** With the equal-phase channel the dark modes decay more slowly as the loss grows, while the slowest mode of independent decay (dashed) decays faster.

and the memory that a readout recovers becomes a quantity that can be engineered in physical systems that process information over time.

2 Model

The reservoir is the driven spin system of Ref. [6]. Five spins ($N = 5$) interact through random couplings J_{ij}, and in each step a random input u between 0 and 1 sets the strength of a transverse field for a time Δt:

$$H(u) = \sum_{i<j} J_{ij} X_i X_j + h \sum_i Z_i + h(1+u) \sum_i X_i, \tag{1}$$

with Pauli operators X_i, Z_i of spin i and a field strength h. To weaken the input at a fixed mean drive, we replace $h(1+u)$ by $h[\frac{3}{2} + \eta(u - \frac{1}{2})]$, where the amplitude η scales how far each input moves the field. The state ρ of the spins follows a Lindblad master equation [40, 41],

$$\dot{\rho} = -i[H(u), \rho] + \sum_k \Big(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \Big), \tag{2}$$

where the Hamiltonian drives the internal dynamics and each jump operator L_k, scaled by one loss rate γ, is one way the environment changes the state, such as removing an excitation or erasing a phase.

A linear readout of 45 features, the single-spin and two-spin expectation values, is trained to estimate the input from τ steps back. The recall C_τ, between 0 and 1, is the squared correlation between the estimate and the input on held-out data, known as the memory function [33]. The linear memory capacity $\sum_\tau C_\tau$ adds up the recall over the delays. One unit is one past input recalled perfectly, and the number of features bounds the capacity [29].

We call the part of the state that depends on the past inputs the stored signal. The memory lifetime t_{mem} is the number of steps over which its variance survives (Appendix A.4). Sections 3.1 to 3.3 and both theorems use only this time scale. One comparison in Sec. 3.4 uses the relaxation time t_{rel} instead, and Table 1 defines all quantities.

Capacities quoted in the text use exact expectation values, the limit of infinitely many shots, and are summed over all delays. Figure 2c and the tables instead show simulated runs summed over the first 300 delays, and Sec. 3.4 counts shots.

All numbers are for five spins with all-to-all couplings. Four statements hold for any number of spins and any Hamiltonian: the total coupling fixes the mean decay rate, the decay out of the dark states falls as $1/\gamma$, an input field with the phases of a collective channel does not couple its dark states, and the two theorems. How the signal scales with the number of spins is open, and reservoirs whose signals concentrate need exponentially many measurements [42].

We compare decay channels at equal total coupling $\mathrm{Tr}\, K = \sum_k \mathrm{Tr}\, L_k^\dagger L_k$ and hold the Hamiltonian fixed to isolate the effect of the form of the dissipation. The generator of Eq. (2) has

4^N modes, patterns of change in the state that each decay at their own rate. Equal total coupling fixes their mean rate for every Hamiltonian, but not the slow modes, which hold a stored signal longest and become slower or faster with the channel (Appendix A.2). Table 2 lists the eight channels.

We discuss four of them (Fig. 1b). With independent decay every spin loses its excitation on its own. With collective decay all spins lose it through one shared channel, with equal phases or with opposite phases on spins 4 and 5, and we refer to the equal-phase and the opposite-phase channel. With dephasing every spin loses its phase. The channels differ in the kernel of the dissipator, the operators that the dissipation alone leaves unchanged, although the Hamiltonian can still turn them or couple them to decaying ones. For independent decay the kernel is the ground state alone. A collective channel cannot empty its dark states, so its kernel contains every operator on the dark subspace. There are $\binom{N}{\lfloor N/2 \rfloor}$ dark states, ten for five spins, and their number grows exponentially with the number of spins. Dephasing erases the coherences between the spin configurations. Every step then leaves the fully mixed state unchanged whatever the input, the driven reservoir relaxes to this state, and dephasing stores nothing.

3 Results

3.1 The decay channel sets the memory lifetime

Increasing the loss separates the channels into two groups (Fig. 2). With independent decay, recall retreats to shorter delays and the linear memory capacity falls steadily (Fig. 2a,c). With the equal-phase channel the capacity first dips and then doubles, and at strong loss inputs fifty steps back are still recalled (Fig. 2b,c). The other channels follow one of the two (Fig. 2c, grey). Exchange and gain/loss follow independent decay, two collective channels follow the equal-phase channel, and pairwise decay follows the opposite-phase channel. The decay rates explain the split, because without dark states the slowest rate grows in proportion to γ and the memory lifetime shrinks, while with dark states the slow rates fall as $1/\gamma$ and the lifetime grows (Fig. 2d).

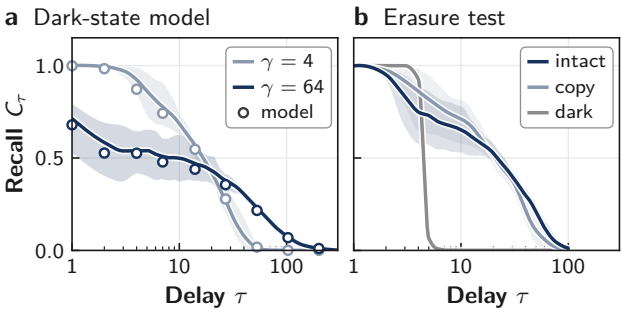


Figure 3: **Dark states hold the memory.** Equal-phase channel. **a** A model of the dark states alone (circles) reproduces the recall of the full reservoir (lines). **b** Erasing what the dark states know about the inputs destroys the memory (dark), while erasing a rotated copy of them leaves most of it (copy).

Zeno protection. Stronger collective decay prolongs the memory through the quantum Zeno effect [25, 43]. In the two-spin picture of Fig. 1c, the dark state $|-\rangle \propto |10\rangle - |01\rangle$ can decay only through the bright state $|+\rangle \propto |10\rangle + |01\rangle$, and strong loss suppresses this escape. Its rate falls as Ω^2/γ, where Ω couples $|+\rangle$ and $|-\rangle$, as the slow mode of an overdamped oscillator settles more slowly the stronger the friction.

For any number of spins the protection follows from perturbation theory in $1/\gamma$ at fixed system size, the adiabatic elimination of the bright states [26, 28]. At strong loss the reservoir follows its Hamiltonian restricted to the dark states, which turns the stored state without making it decay. Decay enters at the next order, when the Hamiltonian briefly lifts a dark state into a bright one. One step is then a rotation plus a loss that falls as $1/\gamma$ (Eq. (A2)), a nearly orthogonal map with a small decay rate, which is the classic recipe for a long memory in a network [33]. Calculations for three to six spins show the same decay out of the dark states with different prefactors, and with the equal-phase channel the capacity also grows with the loss at four and six spins (Appendix E.1).

Where the memory is stored. The dark states hold the memory, as two tests show (Appendix E.1). A model of the ten dark states alone, with no fitted parameter, reproduces the recall of the full reservoir at strong loss (Fig. 3a). Replacing the dark-state components of the state by their time averages after every step, which removes their dependence on the inputs, destroys all memory beyond a few steps, while the same operation on a randomly rotated set of equal size

leaves most of it (Fig. 3b).

Local loss. Residual local loss limits the lifetime. For the equal-phase channel, recalling inputs over T steps requires a ratio of collective to local loss rates of at least $4.2\,T^2$ (Appendix E.6), so that a ratio of 10^4 protects 49 steps. For superconducting qubits coupled to a waveguide, a four-qubit dark state outlives the decay of a single qubit into the waveguide by more than two orders of magnitude [44], and qubit arrays reach a cooperativity of 172 [45].

Dark states protect a stored signal against decay but not against the inputs that follow it. With the opposite-phase channel the dark states are just as protected as with the equal-phase channel, yet the capacity returns only to its weak-loss level (Fig. 2c).

3.2 New inputs mix the stored signal

Each new input turns the stored state by a small angle that depends on the input (Fig. 1d), so every newer input acts as noise on an older one. Over the memory lifetime the stored direction takes a random walk of small turns $\pm\theta$ and spreads by about $\theta\sqrt{t_{\mathrm{mem}}}$. Once the spread reaches about a radian, an old input survives only multiplied by the inputs that followed it, and a linear readout cannot turn such products back into single inputs.

Counting the inputs. A stored signal splits into parts that each depend on a fixed set of past inputs (Appendix A.4), and we denote the number of inputs in the set by D. A function of u_{t-3} alone has $D = 1$, even if it is nonlinear in u_{t-3}, and the product $u_{t-3}u_{t-5}$ has $D = 2$. Linear memory requires $D = 1$. The information processing capacity [29, 46] measures how well a readout reconstructs any function of the past inputs, and sorting its contributions by D separates single inputs from combinations of inputs.

The mixing law. Two quantities decide how many inputs a stored signal combines. The memory lifetime t_{mem} says how long the signal lasts. The mixing fraction s, between 0 and 1, is the part of the signal that one input turns differently from the average input, and for small turns it grows as the square of the turn. Our first theorem states that a part of the stored signal combines on average $s\,t_{\mathrm{mem}}$ inputs. It applies to reservoirs that leak slowly and are mixed by their inputs, and it assumes that one direction of the covariance of the state over input histories outlives all others.

> **Theorem 1** (Mixing law, made precise in Appendix B)**.** Let a reservoir be reset with probability p per step, so that $t_{\mathrm{mem}} \sim 1/(2p)$ as $p \to 0$, and otherwise turned by an input-dependent rotation. Let the average rotation shrink every state, let the rotations leave no stored covariance unchanged on average except the isotropic one, and let one input change the stationary state (conditions (1) to (3) of Appendix B). Pick a part of the stored signal with probability equal to its share of the capacity, and let D be the number of distinct inputs it depends on. As $p \to 0$, D/t_{mem} becomes exponentially distributed with mean s, and
>
> $$\mathbb{E}D = s\,t_{\mathrm{mem}} + O(1).$$

The law follows from a competition between two rates. In each step a fraction s of the stored signal picks up one more input, and a fraction $1/t_{\mathrm{mem}}$ of it is lost. The number of inputs then follows a geometric distribution, which becomes exponential for long memories.

Scope of the law. The proof uses a reservoir with a reset, in any dimension, and nothing in it is quantum. Classical state-affine reservoirs, whose state is updated by an input-dependent linear map plus an offset [47], obey the law under the same conditions. For physical channels without a reset the same proof applies, with the slow decay in place of the reset, if a single direction of the stored covariance is slow. Several slow directions keep distinct mixing scales (Appendix B.3). For five spins, Eq. (A3) gives the distribution of D exactly, and we compare it with the law (Fig. 4c).

Five spins. At a fixed amplitude, $\eta = 1$, one input disturbs the stored signal only slightly, $s \approx 0.02$. For short memories the mean of D is slightly below two, because the reservoir combines the latest inputs within a few steps. As the memory grows, the mean of D follows $s\,t_{\mathrm{mem}}$ to within about two inputs, up to 23 inputs at the longest memory, and the distribution of D approaches the exponential law (Appendix B.3). The product $s\,t_{\mathrm{mem}}$ is the size of the trade-off between memory and nonlinearity [31], and dissipation enters it through both factors.

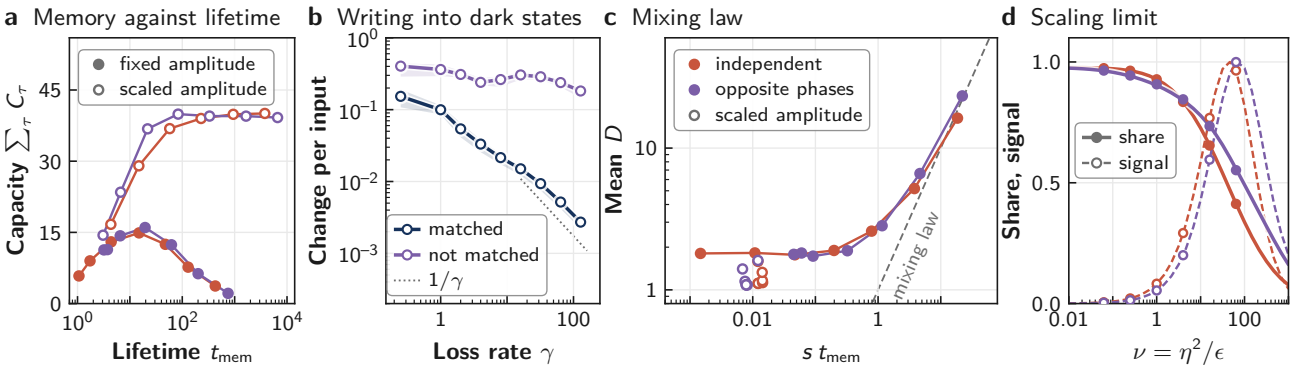


Figure 4: **New inputs mix the stored signal, and a scaled amplitude limits the mixing.** **a** At a fixed input amplitude the capacity, summed over all delays, peaks and then falls as the memory grows, and a scaled amplitude holds it at 40 units. **b** An input that is mode-matched to the equal-phase channel writes less into the dark states as the loss grows, while an input that is not mode-matched keeps writing. **c** The number of inputs mixed into a stored signal follows the mixing law, and a scaled amplitude keeps it near one. **d** The scaling limit (lines) against the exact five-spin calculation (markers), without a fitted parameter. Solid lines and filled markers show the share of single inputs in the capacity, dashed lines and open markers the signal they leave in the features, scaled to its maximum.

At a fixed amplitude, lengthening the memory beyond $s\,t_{\mathrm{mem}} \approx 0.3$ lowers the linear memory capacity (Fig. 4a). With the opposite-phase channel it peaks at 16 units and falls to 2 units while the lifetime grows from 20 to 750 steps, and independent decay behaves in the same way (Appendix E.3). The stored signal has moved into products of inputs, and the share of the capacity carried by single inputs falls from 0.37 to 0.05. Single inputs stay readable only if the mixing fraction shrinks as the lifetime grows. The phases of the decay channel and the amplitude of the input both achieve this.

3.3 Channel phases and input amplitude control the mixing

Channel phases. The phases of a collective channel decide whether an input reaches the dark states. Spin i couples to the channel with the sign $c_i = \pm 1$, and the collective lowering operator $\sum_i c_i \sigma_i^-$ annihilates the dark states. An input field is mode-matched to the channel if its strength on spin i is proportional to c_i, so that it drives the collective mode through which the spins decay. A mode-matched field $\sum_i c_i X_i$ is the lowering operator plus its adjoint and cannot couple two dark states, for any number of spins and any Hamiltonian. It writes into them only through brief excursions into bright states, with a strength that falls as $1/\gamma$. The part of a field that is not mode-matched writes into them directly (Fig. 4b, Appendix E.2).

The uniform input of Eq. (1) is mode-matched to the equal-phase channel. Its mixing fraction shrinks as the loss grows, and this channel reaches twice the capacity of the other two at a fixed amplitude (Table 3). With the opposite-phase channel the same input is not mode-matched. It writes directly into the dark states and mixes them at every loss rate. This is the generic case, because a phase error opens a direct coupling within the dark subspace (Appendix E.2).

Input amplitude. The input amplitude η controls the mixing for every channel. A weak input is known to favour linear memory [30, 31, 35], because it suppresses the products of the latest inputs that the reservoir forms within a few steps. The mixing law adds what a long memory requires. The mixing fraction grows as the square of the amplitude, so $s\,t_{\mathrm{mem}}$ stays fixed only if the amplitude falls as $1/\sqrt{t_{\mathrm{mem}}}$. We refer to this choice as the scaled amplitude and to $\eta = 1$ as the fixed amplitude. Theorem 1 shows that the scaling is necessary, because at a fixed amplitude the share of single inputs vanishes as the memory grows. Our second theorem shows that it is sufficient.

> **Theorem 2** (Scaling limit, made precise in Appendix B.4)**.** Let one step of a reservoir be a rotation, a leak of small rate ϵ and an input-dependent turn of amplitude η, under the decay condition of Appendix B.4. As $\epsilon \to 0$ with $\nu = \eta^2/\epsilon$ fixed, the stored signal and the distribution of D converge to the solution of

a linear problem that depends on the input only through ν. The mean of D stays finite, and single inputs keep a fixed share of the capacity. As $\nu \to \infty$ the limit joins the mixing law, with $s\,t_{\mathrm{mem}}$ proportional to ν.

The scaled amplitude holds ν fixed. The memory lifetime can then grow without limit while the stored signal keeps the same division between single inputs and products of inputs. The leak rate is $\epsilon = \gamma$ for independent decay and $\epsilon = 1/\gamma$ for the dark states, and the limit follows from the Hamiltonian and the jump operators without a fitted parameter. One qubit gives a closed form (Appendix C). For three spins with the opposite-phase channel, interval arithmetic confirms both regimes at strong loss. At a fixed amplitude single inputs lose their share of the capacity, and with an amplitude $\eta_0/\sqrt{\gamma}$ they keep it. The stored signal resides there in coherences between the energy levels of the dark states (Appendix D).

Comparison with five spins. In the five-spin reservoir $\nu = 1$ is an amplitude of $1/\sqrt{\gamma}$ for the opposite-phase channel and of $\sqrt{\gamma}$ for independent decay. The limit reproduces the exact five-spin calculation over three decades of ν, with the share of single inputs agreeing to within 0.01 (Fig. 4d, Appendix E.3). At $\nu = 1$ single inputs carry about 0.9 of the capacity, and the linear memory capacity is 40 units for both channels. The five-spin capacity settles at this level as the lifetime grows (Fig. 4a), up from about 15 units at the best loss rate of a fixed amplitude. Simulated runs on eight reservoirs, with a readout trained on a finite number of inputs and summed over the first 300 delays, reach 35 units (Fig. 2c, Table 3).

The capacities of all values of D add up to the number of features [29], so the gain moves capacity from combinations of inputs to single inputs. The scaled amplitude is the choice for recall. A task that needs products of inputs calls for a larger amplitude, and $s\,t_{\mathrm{mem}}$ gives the number of inputs that a stored signal then combines. A long memory needs no dark states, because weak independent decay with a scaled amplitude reaches one too. Dark states add a memory that stays long under strong dissipation and grows with the loss.

The cost of a weak input. The scaled amplitude is an upper bound on the input and not an optimum. With exact expectation values the capacity keeps rising as the input weakens, towards the number of features (Appendix E.4), because a weaker input mixes less and a fainter signal costs nothing without measurement noise. The lifetime bounds the amplitude only from above, $s\,t_{\mathrm{mem}} \lesssim 1$. Theorem 2 quantifies what a weak input gives up (Fig. 4d). The variance that single inputs leave in the features vanishes in proportion to ν for weak inputs. It is largest where a stored signal combines a few inputs and single inputs carry about half of the capacity (Appendix E.3). The limit depends on the amplitude only through ν, so the amplitude of the largest signal also falls as $1/\sqrt{t_{\mathrm{mem}}}$.

3.4 The stationary state sets the measurable signal

A weaker input leaves a fainter signal. This costs nothing with exact expectation values and many measurements otherwise. Each feature is estimated from n shots per measurement setting and step, and its statistical error must stay below the signal, the variation of the feature with the inputs (Appendix A.5). A memory with a scaled amplitude grows longer without growing larger, because the capacity cannot exceed the number of features, so each delay is recalled more faintly. The capacity recovered from a finite number of shots is the counterpart, for the recall of past inputs, of the resolvable expressive capacity of Ref. [32].

Signal size. The stationary state sets the size of the signal. An input acts on the stationary state ρ_* through the commutator with the input field, and only the part of this change that lies on the slow modes is remembered. Its size is the linear response of the stationary state to the input field (Appendix E.5), in the spirit of the link between memory and optical absorption of Ref. [9].

To compare all eight channels we choose the loss rate of each so that its slow modes relax in 32 steps at the mean input. This relaxation time t_{rel} replaces the memory lifetime in this comparison only, because it does not depend on the input amplitude (Table 1). The variance of the features then follows the linear response over four orders of magnitude for the seven channels that respond (Fig. 5b). The fraction of the exact ca-

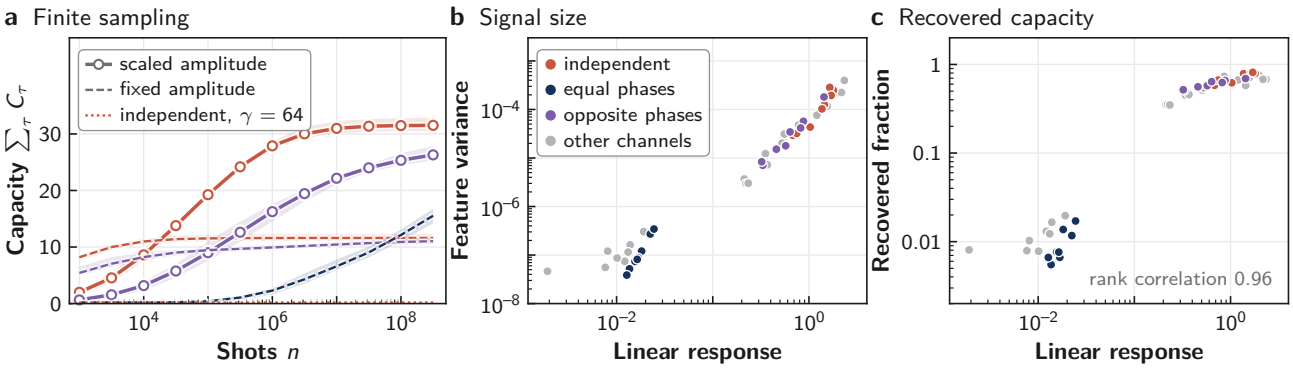


Figure 5: **Finite sampling limits the memory that can be read out. a** Capacity recovered against the number of shots per setting and step with the rewinding protocol, for independent decay at weak loss ($\gamma = 1/32$) and the two collective channels at strong loss ($\gamma = 64$). **b** The variance of the features grows with the linear response of the stationary state to the input field. **c** So does the fraction of the exact capacity that 10^6 shots recover. Points in b and c are single reservoirs. Their loss rates give every channel the same relaxation time, $t_{\mathrm{rel}} = 32$ steps (Table 1), and every shot repeats the full input history.

pacity that 10^6 shots recover follows it as well, with a rank correlation of 0.96 over all reservoirs (Fig. 5c). The equal-phase channel leaves a polarized state that the uniform input moves only through bright states. It has the largest exact capacity of all channels and recovers 1% of it. Reversing the coupling phase of two spins lets the input move the state directly (Sec. 3.3), and the opposite-phase channel recovers 60% of its exact capacity, against 73% for independent decay (Appendix E.5). The exact capacity does not predict what measurements recover. The linear response predicts which fraction of it they do.

Strong loss. The form of the dissipation matters most where the loss is strong (Fig. 5a). At $\gamma = 64$ and a fixed amplitude, 10^4 shots recover 8 units with the opposite-phase channel and nothing above the noise floor with the equal-phase channel or with independent decay (Table 3). A collective channel that is not mode-matched to the input is the right choice when strong loss is given, as for emitters that radiate into a common mode [22–24]. Where the loss can be made weak, independent decay recovers more, 11 units from the same number of shots. A short memory needs few shots, and the scaled amplitude pays off only with more. Its capacity overtakes that of the fixed amplitude between 10^4 and 10^5 shots (Appendix E.5).

Total cost. A measurement destroys the state, so every shot restarts the reservoir and repeats the recent inputs before the measurement, the rewinding protocol [10, 13]. The number of repeated steps grows with the memory lifetime. When every repeated step is counted (Appendix A.5), a budget of 10^9 reservoir steps per output step gives the three channels the same capacity at a fixed amplitude, each at its best loss rate (Table 3). The opposite-phase channel holds at a hundredfold stronger loss what independent decay holds at weak loss. With the scaled amplitude, weak independent decay recovers the most, 34 units against 21 with the opposite-phase channel.

With a finite number of shots a best amplitude exists. It falls as a power of the relaxation time, with a fitted exponent close to the $-1/2$ of the scaled amplitude (Appendix E.4). The lifetime bounds the input amplitude from above, and the measurement budget bounds it from below.

4 Conclusion

A useful quantum reservoir must keep a readable signal of the past while it accepts new inputs. We have shown that the form of the dissipation sets three properties of this memory. Its lifetime depends on the decay channel, because independent decay shortens the memory as the loss grows, whereas strong collective decay prolongs it by holding the stored signal in dark states through the quantum Zeno effect.

Writing is limited by mixing, because a longer memory leaves more time for later inputs to mix with the stored signal. The mixing law gives the number of inputs that mix, and the scaling limit shows that an amplitude that falls as the inverse square root of the lifetime keeps this number fixed.

In the spin reservoirs studied, the linear memory capacity then rises from about 15 to 40 of 45 units with exact expectation values.

The signal must be strong enough to resolve, and the equal-phase channel holds the largest memory and leaves the faintest signal. Where the loss can be made weak, independent decay recovers the most because its signals are stronger.

The environment, the input and the readout have to be designed together, and three rules summarize the design:

(1) *Lifetime.* Collective decay protects a memory in dark states. Stronger loss then makes it last longer, as long as local loss stays weak.

(2) *Writing.* The input must reach the dark states, and an input with the phases of the decay channel reaches them ever more weakly as the loss grows. The longer the memory lasts, the weaker each input must be, or new inputs mix with the old ones.

(3) *Signal.* What can be read out depends on how strongly the inputs move the stationary state. At strong loss a collective channel whose phases differ from those of the input recovers the most.

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Data availability

The code, the data behind every figure, and the derivation and certificates of the three-spin limits are at https://github.com/eybmits/what-a-dissipative-quantum-system-remembers.

Author contributions

M.B. contributed to conception, methodology, software, analysis, visualization and writing, I.F. and J.W. to methodology, interpretation and editing, and C.L.-P. and J.S. to supervision and conceptual development. OpenAI Codex and Claude Code (Anthropic) assisted with manuscript preparation, checks of derivations and references, and simulation, verification and plotting code. The authors reviewed the outputs and take responsibility for the final content.

Table 1: Time scales and other quantities. Times are in input steps of length Δt. The theorems use only t_{mem}.

Time scales	
t_{mem}	memory lifetime: decay time of the variance of the stored signal (Appendix A.4). Used in Secs. 3.1 to 3.3.
t_{rel}	relaxation time: decay time of the slow modes at the mean input, independent of the input amplitude (Appendix A.5). Used to match the channels in Fig. 5b,c and Table 4.
t_{f}	recall time: decay time of the response to one input, with $1/t_{\mathrm{f}} = 1/t_{\mathrm{rel}} + O(\eta^2)$ (Appendix A.3). Used in Appendix E only.
Other quantities	
D	number of distinct past inputs that a part of the stored signal depends on
s	mixing fraction: the part of a stored signal that one input turns differently from the average input
ν	squared input amplitude per leak rate, η^2/ϵ, which the scaled amplitude holds fixed (Appendix B.4)
τ_{90}	rewinding window: the first τ_{90} delays hold 90% of the exact recall (Appendix A.5)

A Methods

This appendix specifies the five-spin simulations and defines the quantities of the main text, which Table 1 lists.

A.1 Reservoir, readout and capacity

The Hamiltonian of Eq. (1) uses $h = 1/2$ and couplings J_{ij} drawn uniformly from $[-1, 1]$, so the transverse field runs from h to $2h$. Each input u is drawn uniformly from $[0, 1]$ and held for a step of length $\Delta t = 1/2$. We integrate Eq. (2) in the 1024-dimensional basis of Pauli operators and evaluate the step map $E(u) = e^{\Delta t \mathcal{L}(u)}$ by Chebyshev interpolation in u at ten nodes. Each reservoir draws its couplings and its input stream from one seed, and every channel uses the same eight seeds. After a washout of $\max(800, 60\gamma)$ steps, at least four times the recall time, we record the 45 features, the expectation values of X_i, Y_i, Z_i and of X_iX_j, Y_iY_j, Z_iZ_j, for 2000 training, 400 validation and 1600 test inputs. The readout is a least-squares fit

Table 2: Decay channels at five spins, each with $\operatorname{Tr} K = 80\gamma$, and the number of operators that the dissipator leaves unchanged. The operators $\sigma_i^{\mp}$ lower and raise spin i, with $i < j$ for pairwise decay, $i \neq j$ for exchange and $c = (1, 1, 1, -1, -1)$.

channel	jump operators	kernel dimension
independent	$\sqrt{\gamma}\,\sigma_i^-$	1
equal phases	$\sqrt{\gamma}\sum_i \sigma_i^-$	100
opposite phases	$\sqrt{\gamma}\sum_i c_i\sigma_i^-$	100
two channels	$\sqrt{\gamma}\sum_{i\leq 3}\sigma_i^-$, $\sqrt{\gamma}\sum_{i>3}\sigma_i^-$	36
pairwise	$\sqrt{\gamma}\,\sigma_i^-\sigma_j^-$	36
exchange	$\sqrt{0.4\gamma}\,\sigma_i^-$, $\sqrt{0.3\gamma}\,\sigma_i^-\sigma_j^+$	1
gain/loss	$\sqrt{\gamma/1.3}\,\sigma_i^-$, $\sqrt{0.3\gamma/1.3}\,\sigma_i^+$	1
dephasing	$\sqrt{\gamma/2}\,Z_i$	32

with an intercept on the training and validation inputs, and C_τ is the squared correlation of its estimate with $u_{t-\tau}$ on the test inputs. In these simulated runs the linear memory capacity sums C_τ over delays 1 to 300, and the exact moments of Appendix A.4 give the sum over all delays. A delay without any recall reads about 1/1600 on the test inputs, so the sum has a floor of about 0.2 units. Figures show the mean of the eight reservoirs, with bands from the 16th to the 84th percentile.

A.2 Decay channels and equal coupling

Table 2 lists the jump operators of the eight decay channels. Their weights give every channel the same total coupling $\operatorname{Tr} K = N2^{N-1}\gamma = 80\gamma$, which fixes the mean decay rate. For a traceless jump L the superoperator trace of its dissipator is $-2^N \operatorname{Tr} L^\dagger L$, while the commutator with any Hamiltonian is traceless, so the 4^N eigenvalues λ of the generator obey

$$\frac{1}{4^N}\sum \operatorname{Re}\lambda = -\frac{\operatorname{Tr} K}{2^N} \tag{A1}$$

for every channel and every Hamiltonian. The kernel of the dissipator is larger than the set of dark states: the ten dark states of a collective channel carry $10 \times 10 = 100$ operators.

A.3 Slow modes, recall time and the effective generator

The input-averaged step $\bar{E} = \mathbb{E}_u E(u)$ is computed by six-point Gauss–Legendre quadrature. Its eigenvalues λ_j give decay rates $\kappa_j = -\ln|\lambda_j|/\Delta t$. If the kernel of the dissipator has dimension n, the slow modes are the $n-1$ slowest decaying ones besides the stationary state, and the recall time is $t_{\mathrm{f}} = 1/(\Delta t\,\bar{\kappa})$, with $\bar{\kappa}$ their mean rate. For $n = 1$ we use the slowest mode. The averaged step carries the part of a stored signal that still depends on a single input. Its mean rate adds decay and mixing to second order in the amplitude, $\bar{\kappa} = \bar{\kappa}_0 + c\,\eta^2 + O(\eta^4)$, because the inputs are distributed symmetrically about their mean and the term of $\bar{E}$ linear in η vanishes. Appendix E.2 tests this sum numerically. Figure 2d shows instead the decay rates $-\operatorname{Re}\lambda$ of the generator at the mean input, $u = 1/2$. With a collective channel we call the $n-1$ slowest of them dark and all faster ones bright.

Let $\mathcal{D}$ be the unit-rate dissipator, $\mathcal{P} = \lim_{t\to\infty} e^{t\mathcal{D}}$ the projector onto its kernel and $\mathcal{S} = (\mathcal{D}+\mathcal{P})^{-1} - \mathcal{P}$ its inverse on the decaying operators. At strong loss, adiabatic elimination of the decaying operators to second order [26, 28] gives the effective generator on the kernel,

$$\mathcal{G}(u) = \mathcal{P}\mathcal{H}(u)\mathcal{P} - \frac{1}{\gamma}\,\mathcal{P}\mathcal{H}(u)\mathcal{S}\mathcal{H}(u)\mathcal{P}, \tag{A2}$$

with $\mathcal{H} = -i[H, \cdot\,]$. The first term is the Hamiltonian restricted to the kernel, which turns its operators, and the second is the loss through brief excursions into bright states.

The scaled amplitude is $\eta = \min(1, \gamma^{-1/2})$ for the opposite-phase channel and $\eta = \min(1, \gamma^{1/2})$ for independent decay, which is $\nu = 1$ in Appendix B.4.

A.4 Number of inputs mixed

The order D of a stored signal is the number of distinct past inputs it depends on (Sec. 3.2). Precisely, for independent inputs, every function of the past inputs with finite variance splits uniquely into orthogonal parts, each of which depends on a fixed finite set of inputs, the Hoeffding decomposition [48]. The order of a part is the size of its set, not a polynomial degree. For signs $u_t = \pm 1$ the parts are the products $\prod_{j\in J} u_{t-j}$, and for continuous inputs one uses orthogonal polynomials in each input. We collect the traceless coefficients of the state in an orthonormal operator basis in a vector r, on which quantum channels act affinely [49, 50], $r_t = A(u_t)r_{t-1} + b(u_t)$. All means and covariances are taken over input histories. Let $\bar{A} = \mathbb{E}A$ be the mean step, m the stationary mean and $\xi = A(u)m + b(u) - m$ the innovation, the

change that one input writes, with covariance V. With $\mathcal{B}X = \bar{A}X\bar{A}^\mathsf{T}$, $\mathcal{T}X = \mathbb{E}[AXA^\mathsf{T}]$ and $\mathcal{K} = \mathcal{T} - \mathcal{B}$, the covariance G_k carried by products of order k obeys

$$G_1 = (1-\mathcal{B})^{-1}V, \qquad G_{k+1} = (1-\mathcal{B})^{-1}\mathcal{K}G_k. \tag{A3}$$

The innovation adds the current input alone. The mean step carries a stored product along without adding an input. The fluctuating part $\mathcal{K}$ of the step multiplies it by the current input, which raises its order by one. The G_k add up to the stationary covariance Σ. A readout of features $y = Mr$ has the capacity $\mathrm{Tr}[(M\Sigma M^\mathsf{T})^+ MG_kM^\mathsf{T}]$ of order k, and these capacities add up to the rank of the readout [29]. Divided by their sum, they give the share of each order, the distribution of D in Theorem 1. For one observable with coordinates a the share of order k is $a^\mathsf{T}G_ka/a^\mathsf{T}\Sigma a$, and the share of single past inputs is $a^\mathsf{T}\bar{A}G_1\bar{A}^\mathsf{T}a/a^\mathsf{T}\Sigma a$.

The memory lifetime and the mixing fraction come from the slowest mode of the covariance step $\mathcal{T}$: its eigenvalue is $1-1/t_{\mathrm{mem}}$, and with its right eigenvector Q and left eigenvector ℓ the mixing fraction is $s = \ell(\mathcal{K}Q)/\ell(Q)$, as in Appendix B.3. For five spins we evaluate Eq. (A3) on the 1023 traceless Pauli coordinates (Appendix E.3).

A.5 Finite sampling and the measurement budget

A quantum measurement gives a random outcome, so every feature is an average over n shots per measurement setting and step. We use three settings, with all spins read along X, Y or Z. For each step we form the outcome distributions of the three settings and draw n shots from each. The readout is fitted as above, with a ridge chosen from fourteen values between 0 and 10^4 on the validation inputs. In Sec. 3.4 the distributions come from the rewinding protocol [10]: for each output step the reservoir is restarted in $|00000\rangle$ and the last τ_{90} inputs are repeated, where the first τ_{90} delays hold 90% of the exact recall. We count the budget B in reservoir steps, shots times repeated steps summed over the three settings, per output step, so that $n = B/(3\tau_{90})$, and we also run $n = 10^3$ to $10^{8.5}$ in half decades. Only Fig. 5b,c and Table 4 use the state with the full history. Their loss rates are matched to a relaxation time $t_{\mathrm{rel}} = 1/(\Delta t\,\bar{\kappa}_0)$ of 32 input steps, where $\bar{\kappa}_0$ is the mean decay rate of the slow modes of the generator at the mean input, without the mixing by the inputs (Appendix E.5).

B The mixing law and the scaling limit

This appendix proves the two theorems and extends the mixing law to physical channels without a reset.

B.1 Setting and statement

We describe the reservoir by the n traceless coordinates r of its state in an orthonormal basis of operators, the Bloch vector for a qubit. A unitary acts on r as a rotation. In each step the reservoir is reset to a fixed state with a small probability p and is otherwise rotated by $O(u)$:

$$r_t = (1-p)\,O(u_t)\,r_{t-1} + p\,b. \tag{B1}$$

Here b represents the reset state. Equation (B1) averages over whether the reset occurred, so the only remaining randomness is the input history, with independent and identically distributed inputs u_t. A stored signal loses a fraction $1-(1-p)^2 = 2p-p^2$ of its variance per step, so its memory lifetime, defined by the slow eigenvalue $1-1/t_{\mathrm{mem}}$ of the covariance step (Appendix A.4), is $t_{\mathrm{mem}} = 1/(2p-p^2)$, or $1/(2p)$ to leading order. Three conditions make the law hold:

(1) *The average step forgets.* The average rotation $A = \mathbb{E}O$ shrinks every vector: its spectral radius is below one.

(2) *The inputs mix.* The only covariance matrices that the random rotations leave unchanged on average, $\mathbb{E}[OXO^\mathsf{T}] = X$, are multiples of the identity.

(3) *The input writes.* One input changes the state: $\mathbb{E}\|(O-A)\mu\|^2 > 0$ with $\mu = (1-A)^{-1}b$.

The first condition means that only products of inputs can be stored for long, the second that a stored signal spreads evenly over all directions, and the third that there is something to store. The mixing fraction

$$s = 1 - \frac{\|A\|_F^2}{n} \tag{B2}$$

says how strongly one input turns the stored state. For a unit vector x of random orientation, $\mathbb{E}\|(O-$

$A)x\|^2 = s$: one input turns a fraction s of a stored signal away from the average rotation. The fraction lies between 0 and 1, the first condition makes it positive, and for small turns θ it grows as θ^2.

Mixing law, precise form. Assume conditions (1)–(3) and take any fixed linear readout. Pick a part of the stored signal with probability equal to its share of the readout's capacity, and let D be its order, the number of distinct inputs it depends on. As $p \to 0$, the ratio D/t_{mem} becomes exponentially distributed with mean s, and

$$\mathbb{E}D = s\, t_{\text{mem}} + O(1).$$

Equivalently, products of more than $x\, t_{\text{mem}}$ inputs carry the share $e^{-x/s}$ of the capacity.

B.2 Proof

The proof makes the competition of Sec. 3.2 exact: on average $1/s$ steps pass before the next input joins a product, which meanwhile loses a fraction $1/(s\, t_{\text{mem}})$ of its weight. The proof works with covariance matrices. Appendix A.4 splits the stationary covariance Σ of the state into the parts G_k carried by products of k inputs and gives their recursion, Eq. (A3): the mean step $\mathcal{B}$ carries a stored product along, and the fluctuating part $\mathcal{K}$ of the step adds one input to it. In the reset model the mean step is $\mathcal{B}X = (1-p)^2 AXA^\mathsf{T}$, the full step $\mathcal{T}X = (1-p)^2\,\mathbb{E}[OXO^\mathsf{T}]$, and $\mathcal{K} = \mathcal{T} - \mathcal{B}$. We write $\mathcal{B}_0$, $\mathcal{T}_0$ and $\mathcal{K}_0$ for the same maps at $p = 0$.

Step 1: the stored signal is small and spread evenly. The stationary mean is $m = p\,[1-(1-p)A]^{-1}b = p\mu + O(p^2)$, so one input writes the change $(1-p)(O-A)m$, of size p. Its covariance V has the trace $p^2 v + O(p^3)$ with $v = \mathbb{E}\|(O-A)\mu\|^2$, which is positive by condition (3). The random rotations preserve the trace of a covariance and, by condition (2), leave only the identity unchanged. Since they cannot enlarge a covariance, the identity is a simple fixed point with the projector $X \mapsto (\operatorname{Tr} X/n)\,1$. On every other direction $(1-\mathcal{T})^{-1}$ stays bounded as $p \to 0$, while on the identity it equals $1/[1-(1-p)^2] \approx 1/(2p)$. The written covariance therefore piles up along the identity:

$$\Sigma = (1-\mathcal{T})^{-1}V = \frac{pv}{2n}\,1 + O(p^2). \tag{B3}$$

Step 2: few inputs carry a vanishing share. The mean step forgets by condition (1), so $(1-\mathcal{B})^{-1}$ stays bounded as $p \to 0$. The recursion then makes $G_1 = (1-\mathcal{B})^{-1}V$ of order p^2, and the same holds for every G_k with fixed k. Measured against Σ, which is of order p and has full rank, any fixed set of orders carries a share of order p of the capacity of any readout. A long memory consists of products of many inputs.

Step 3: counting the inputs. Call $\mathcal{F} = (1-\mathcal{B})^{-1}\mathcal{K}$ the order transfer, so that $G_{k+1} = \mathcal{F}G_k$. At $p = 0$ it leaves the identity unchanged, because $\mathcal{K}_0 1 = 1 - AA^\mathsf{T} = (1-\mathcal{B}_0)1$. Its matching left eigenvector is the weight

$$\ell(X) = \operatorname{Tr}[(1 - A^\mathsf{T}A)X], \qquad \ell(1) = ns, \tag{B4}$$

since $\ell(\mathcal{F}_0 X) = \operatorname{Tr}\mathcal{K}_0 X = \ell(X)$ when the rotations preserve the trace. For small p the eigenvalue one moves to

$$1 - \frac{2}{s}\,p + O(p^2)\,: \tag{B5}$$

each step up in order costs a fraction $2p/s = 1/(s\, t_{\text{mem}})$ of the weight, as in that competition. To see this, differentiate $(1-\mathcal{B})\mathcal{F} = \mathcal{K}$ at $p = 0$ and apply it to the identity, which gives $(1-\mathcal{B}_0)\mathcal{F}_0' 1 = -2 \cdot 1$. Its trace is the weight $\ell(\mathcal{F}_0' 1) = -2n$, and dividing by $\ell(1) = ns$ gives the rate $2/s$.

To turn the eigenvalue into a distribution, weigh order k with $e^{-\lambda p k}$. Summing the recursion gives $\sum_k z^k G_k = z(1 - z\mathcal{F})^{-1}G_1$, and as $z = e^{-\lambda p}$ approaches one, only the eigenvalue of Eq. (B5) makes the inverse large. With $\ell(G_1) = p^2 v + O(p^3)$ and $\beta = 2/s$,

$$\sum_{k\geq 1} e^{-\lambda p k} G_k = \frac{pv}{2n}\,\frac{\beta}{\lambda+\beta}\,1 + O(p^2). \tag{B6}$$

Divided by Eq. (B3), this is $\beta/(\lambda+\beta)$, the Laplace transform of an exponential distribution of pD with rate β. So $D/t_{\text{mem}} = 2pD\,[1 + O(p)]$ is exponential with mean s in the limit. Because both covariances are multiples of the identity to leading order, the same holds for the share of the capacity of any readout. Finally, $\sum_k kG_k = (1-\mathcal{F})^{-1}\Sigma = \Sigma/(\beta p) + O(p)$ gives $\mathbb{E}D = s\, t_{\text{mem}} + O(1)$.

Two technical points. On symmetric matrices, where covariances live, the eigenvalue one of $\mathcal{F}_0$ is simple: a fixed point satisfies $\mathcal{T}_0 X = X$ and is therefore a multiple of the identity, and $\ell(1) > 0$ rules out a Jordan chain. For $p > 0$ every eigenvalue of $\mathcal{F}$ lies inside the unit circle, by a Perron–Frobenius argument: $\mathcal{F}$ maps positive matrices to positive matrices, and so does $(1 - \mathcal{F})^{-1} = 1 + (1 - \mathcal{T})^{-1}\mathcal{K}$. The eigenvalues of $\mathcal{F}_0$ therefore lie in the closed unit disk, only the simple one equals one, and only the eigenvalue of Eq. (B5) makes the sum diverge.

B.3 Physical channels without a reset

A physical channel does not reset to a prescribed state, but the proof used the reset only through three facts: one covariance direction is slow, it leaks at a small rate, and the mean step forgets everything else. The same three steps therefore prove the law for general channels.

Setting. One step is $r_t = A_u(\epsilon) r_{t-1} + b_u(\epsilon)$ on the coordinates that the inputs can reach, with a small leak ϵ, and $\mathcal{B}$, $\mathcal{T}$ and $\mathcal{K}$ are formed from A_u as before. We assume:

(i) A_u and b_u depend smoothly on ϵ in mean square, and b_u vanishes at $\epsilon = 0$.

(ii) The mean step forgets at $\epsilon = 0$: $\mathbb{E} A_u(0)$ has spectral radius below one.

(iii) The channel is mean-square stable for small $\epsilon > 0$: the covariance step $\mathcal{T}$ has spectral radius below one.

(iv) One covariance direction is slow: $\mathcal{T}$ has a simple eigenvalue $1 - \kappa\epsilon + O(\epsilon^2)$ with $\kappa > 0$ and no other eigenvalue near one. At $\epsilon = 0$ its right eigenvector is a positive matrix Q, and its left eigenvector is a weight ℓ that is nonnegative on positive matrices, with $\ell(Q) = 1$.

(v) The input writes along it: the change written by one input, of order ϵ, has covariance $\epsilon^2 V_* + O(\epsilon^3)$ with $\ell(V_*) > 0$.

The reset model is the case $\epsilon = p$, $\kappa = 2$, $Q = 1$ and $\ell = \mathrm{Tr}\,/n$. Condition (iv) concerns the covariance that the random steps carry, not the number of slow modes: at a fixed amplitude the inputs turn the 99 slow dark modes of the opposite-phase channel differently in each step and spread a stored signal over all of them faster than it leaks, so that one covariance direction stays slow. Without (iv) one observable shows several order scales: two reset sectors with mixing fractions $1/4$ and $3/4$, selected by a classical flag, give products of more than $x\, t_{\mathrm{mem}}$ inputs the share $w\, e^{-4x} + (1-w)\, e^{-4x/3}$ with $w = 0.82$, which exact moments reproduce.

> **Mixing law without a reset.** Assume (i)–(v) and take a fixed readout $y = Mr$ whose projected slow covariance MQM^{T} has full rank. Then the mixing fraction $s = \ell(\mathcal{K}_0 Q)$ is positive and at most one, and ϵD becomes exponentially distributed with rate κ/s. The mean order is $s/(\kappa\epsilon) + O(1)$: the mixing fraction times the memory lifetime $1/(\kappa\epsilon)$.

Proof. Steps 1 and 2 carry over. The stationary mean is of order ϵ by (i) and (ii), and the slow eigenvalue piles the written covariance up along Q,

$$\Sigma = \epsilon\, \frac{\ell(V_*)}{\kappa}\, Q + O(\epsilon^2),$$

and every fixed order carries a covariance of order ϵ^2. Here the readout condition enters: the shares of the orders in the capacity of the readout follow from Σ only if MQM^{T} has full rank. For one observable a this requires $a^{\mathsf{T}} Q a > 0$. In step 3, $\mathcal{F}_0$ has the same fixed points as $\mathcal{T}_0$, so Q spans them, and $\ell \circ (1 - \mathcal{B}_0)$ is its left eigenvector, with value $\ell(\mathcal{K}_0 Q) = s$ on Q. Differentiating $(1 - \mathcal{B})\mathcal{F} = \mathcal{K}$ at $\epsilon = 0$ and applying it to Q gives $(1 - \mathcal{B}_0)\mathcal{F}_0' Q = \mathcal{T}_0' Q$, whose weight $\ell(\mathcal{T}_0' Q) = -\kappa$ is the rate of the slow eigenvalue of $\mathcal{T}$. The order eigenvalue is therefore $1 - (\kappa/s)\epsilon + O(\epsilon^2)$, and the Laplace transform follows as before. It remains to show that s is positive, which also makes the eigenvalue simple. Because $\mathcal{K}_0$ maps positive matrices to positive matrices and ℓ is nonnegative on them, $\ell(\mathcal{B}_0 Y) \le \ell(\mathcal{T}_0 Y) = \ell(Y)$ for every positive Y, so $0 \le \ell(\mathcal{B}_0^j \mathcal{K}_0 Q) \le s \le 1$ for every j. If s vanished, the series $\mathcal{F}_0 = \sum_j \mathcal{B}_0^j \mathcal{K}_0$ would give $\ell(\mathcal{F}_0 Q) = 0$, which contradicts $\mathcal{F}_0 Q = Q$ and $\ell(Q) = 1$.

For three spins the certificates of Appendix D check these conditions, and for five spins we check the law itself.

Numerical check. We compared the law with exact moments for 24 reset reservoirs and readouts in dimensions two to four. The largest deviation from the law shrinks in proportion to p, from 0.08 at $p = 3\%$ to 0.003 at $p = 0.1\%$. For five

spins the share of the capacity in products of more than K inputs approaches $e^{-K/(s\,t_{\mathrm{mem}})}$: its largest deviation, over K and the reservoirs, falls from 0.33 and 0.43 at $s\,t_{\mathrm{mem}} \approx 1$ to 0.07 and 0.14 at $s\,t_{\mathrm{mem}} \approx 20$, with the opposite-phase channel and with independent decay.

B.4 The scaling limit

Setting. The amplitude now falls with the leak, $\eta^2 = \nu\epsilon$ at fixed ν, and uniformly in the input

$$\begin{aligned} A_u &= O_0\big[1 + \eta\, a_1(u) + \eta^2 a_2(u) - \epsilon Y\big] + o(\epsilon), \\ b_u &= \epsilon\, O_0 c_0 + o(\epsilon), \end{aligned} \tag{B7}$$

with a rotation O_0, $\mathbb{E}a_1 = 0$, and a step that is a rotation without the leak, so that $a_1^{\mathsf{T}} = -a_1$ and $a_2 + a_2^{\mathsf{T}} = a_1^2$. Weak independent decay has this form with $\epsilon = \gamma$, and the effective generator of Eq. (A2) has it on the dark coordinates with $\epsilon = 1/\gamma$: O_0 is the step of the Hamiltonian at the mean input, a_1 and a_2 are its response to the input field, and Y and c_0 the leak. Let Π_0 project onto the vectors that O_0 leaves fixed and Π onto the symmetric matrices that commute with O_0, the slow covariances. For d energy levels with distinct transition frequencies these are the covariances of the $d-1$ populations and one variance for each coherence, $d(d-1)$ numbers. On them, with $\bar{a}_2 = \mathbb{E}a_2$,

$$\begin{aligned} \mathsf{L}X &= \Pi(YX + XY^{\mathsf{T}}), \\ \mathsf{S}X &= -\Pi(\bar{a}_2 X + X\bar{a}_2^{\mathsf{T}}), \\ \mathsf{K}X &= \Pi\,\mathbb{E}[a_1 X a_1^{\mathsf{T}}] \end{aligned} \tag{B8}$$

are the leak, what the inputs take from a stored product on average, and what they pass on to the next order, with $\operatorname{Tr}\mathsf{S}X = \operatorname{Tr}\mathsf{K}X$.

Scaling limit, precise form. Let every slow covariance decay: the eigenvalues of $\mathsf{L}+\nu(\mathsf{S}-\mathsf{K})$ have positive real parts. As $\epsilon \to 0$ at fixed ν:

(a) The stationary mean converges to the vector m_0 fixed by O_0 with $\Pi_0(Y - \nu\bar{a}_2)m_0 = \Pi_0 c_0$.

(b) With $V_1 = \mathbb{E}[a_1 m_0 m_0^{\mathsf{T}} a_1^{\mathsf{T}}]$ and $\mathsf{F}_\nu = \nu(\mathsf{L}+\nu\mathsf{S})^{-1}\mathsf{K}$,

$$\begin{aligned} \Sigma \to \Sigma_\nu &= \nu\,[\mathsf{L} + \nu(\mathsf{S}-\mathsf{K})]^{-1}\Pi V_1, \\ G_k \to \mathsf{F}_\nu^{k-1}\,&\nu\,(\mathsf{L}+\nu\mathsf{S})^{-1}\Pi V_1. \end{aligned}$$

(c) F_ν has a spectral radius $\varrho < 1$. For a readout $y = Mr$ with $M\Sigma_\nu M^{\mathsf{T}}$ of full rank, the distribution of D converges to that of (b), with a finite mean and the share of $k = 1$ for single inputs.

(d) As $\nu \to 0$, Σ_ν and ϱ vanish in proportion to ν. As $\nu \to \infty$, if the identity is the only slow covariance that the inputs leave unchanged, Σ_ν vanishes as $1/\nu$ and $\varrho/(1-\varrho) = \nu\,\mathbb{E}\|a_1\|_F^2/(2\operatorname{Tr}Y) + O(1)$, the $s\,t_{\mathrm{mem}}$ of the reset model.

Proof. Over a lifetime O_0 turns many times, so only what turns with it accumulates, and on that the leak and the inputs act at the same order ϵ. This is the weak-coupling limit of open quantum systems [51], taken for the covariance over input histories. It rests on a lemma: let R be orthogonal, P the projector onto the vectors it leaves fixed and $M_\epsilon = 1 - R - \epsilon RN + o(\epsilon)$. If $\Gamma = -PNP$ is invertible on the range of P, then $\epsilon M_\epsilon^{-1} \to \Gamma^{-1}P$. Indeed $PR = RP = P$, so in the blocks of P and $1-P$ the first diagonal block of M_ϵ is $\epsilon\Gamma + o(\epsilon)$, the off-diagonal blocks are of order ϵ, and the second tends to $1-R$, which is invertible there.

With $R = O_0$ the lemma gives (a), since $1-\bar{A} = 1 - O_0 - \epsilon O_0(\nu\bar{a}_2 - Y) + o(\epsilon)$. On symmetric matrices $\mathcal{R}_0 X = O_0 X O_0^{\mathsf{T}}$ is orthogonal, and Π projects onto its fixed points. With primes for the maps of Eq. (B8) without Π, Eq. (A3) has $\mathcal{B} = \mathcal{R}_0[1-\epsilon(\mathsf{L}'+\nu\mathsf{S}')]$, $\mathcal{K} = \epsilon\nu\,\mathcal{R}_0\mathsf{K}'$ and, because one input writes $\eta\,O_0 a_1 m_0$, $V = \epsilon\nu\,\mathcal{R}_0 V_1$, each up to $o(\epsilon)$. The lemma gives $\epsilon(1-\mathcal{B})^{-1} \to (\mathsf{L}+\nu\mathsf{S})^{-1}\Pi$ and $\epsilon(1-\mathcal{T})^{-1} \to [\mathsf{L}+\nu(\mathsf{S}-\mathsf{K})]^{-1}\Pi$, hence (b), and the order transfer $(1-\mathcal{B})^{-1}\mathcal{K}$ converges to a map whose nonzero eigenvalues are those of F_ν. The orders then fall geometrically at a common rate for small ϵ, which gives (c) once $\varrho < 1$.

The semigroups generated by $-(\mathsf{L}+\nu\mathsf{S})$ and $-[\mathsf{L}+\nu(\mathsf{S}-\mathsf{K})]$ keep positive matrices positive, as projections with the positive map Π of $X \mapsto e^{tW}Xe^{tW^{\mathsf{T}}}$, $W = \nu\bar{a}_2 - Y$, without and with the positive map $\nu\mathsf{K}'$ added to its generator, and the first is bounded by the second. The second decays by assumption, so both generators have positive inverses, and $(1-\mathsf{F}_\nu)^{-1} = 1 + \nu[\mathsf{L}+\nu(\mathsf{S}-\mathsf{K})]^{-1}\mathsf{K}$ is positive, which gives $\varrho < 1$ as in Appendix B.2. On covariances of fixed vectors $\mathsf{L}+\nu\mathsf{S}$ acts as $X \mapsto W_0 X + XW_0^{\mathsf{T}}$ with $W_0 = \Pi_0(Y - \nu\bar{a}_2)\Pi_0$, so W_0 is invertible and m_0 exists.

For (d), (b) is linear in ν to first order. For large ν, $\mathsf{F}_\nu \to \mathsf{S}^{-1}\mathsf{K}$, which leaves the identity unchanged because $\mathsf{S}1 = \mathsf{K}1$, with the left eigenvector $X \mapsto \operatorname{Tr}\mathsf{S}X$. To first order in $1/\nu$ its eigenvalue moves to $1 - 2\operatorname{Tr}Y/(\nu\,\mathbb{E}\|a_1\|_F^2)$. The mean m_0 falls as $1/\nu$, V_1 as $1/\nu^2$, and Σ_ν, which piles up along the identity as in Eq. (B3), as $1/\nu$. In the reset model $Y = 1$ and $s = \eta^2\,\mathbb{E}\|a_1\|_F^2/n$ to leading order.

C One qubit in closed form

A qubit relaxes towards $+X$ at a rate ϵ, through the jump $\sqrt{\epsilon}\,|+x\rangle\langle -x|$, is turned about Z by $u\theta$ per step by the inputs $u = \pm 1$, and is read along Y. The averaged step keeps the axes X and Y apart and the input-dependent part swaps them, so Y carries only odd orders: order $2j+1$ has the share $(1-\chi)\chi^j$ of its capacity, where χ is the factor by which the recursion (A3) returns covariance from Y through X to Y, and $\mathbb{E}D = (1+\chi)/(1-\chi)$. At a fixed angle with $\sin\theta \neq 0$, $1 - \chi = 3\epsilon/\sin^2\theta + O(\epsilon^2)$ as $\epsilon \to 0$: ϵD becomes exponentially distributed with rate $3/(2\sin^2\theta)$, and the share of single inputs vanishes. If instead $\theta^2/\epsilon \to \nu$, Appendix B.4 applies with $O_0 = 1$ and gives

$$\chi \to \frac{\nu^2}{(\nu+2)(\nu+1)}, \qquad \operatorname{Var} Y \to \frac{4\nu}{(\nu+2)(3\nu+2)}. \tag{C1}$$

Single past inputs keep the share $1-\chi$. The signal is largest at $\nu = 2/\sqrt{3}$, where single inputs carry 80% of the capacity, and vanishes as $4/(3\nu)$ for $\nu \gg 1$ and as ν for $\nu \ll 1$. Direct propagation of the density matrix reproduces these expressions.

D Three spins at strong loss

What is certified. Take three spins with a collective channel of opposite phases, Eq. (D1), four Hamiltonians H1–H4 with couplings drawn from fixed seeds, and every Hamiltonian whose three couplings lie in a small box around theirs. As γ grows:

- At a fixed input amplitude, the order of the memory of X_1, divided by γ, becomes exponentially distributed, and the signal in X_1 falls as $1/\gamma$ (boxes of half-width 1/32).
- With the amplitude scaled as $8/\sqrt{\gamma}$, single past inputs keep a finite share of the capacity of X_1, and the range of delays that holds 90% of their recall grows in proportion to γ (boxes of half-width 1/512).

Model and check. The three spins evolve under

$$H_\eta(u) = \sum_{i<j} J_{ij} X_i X_j + \tfrac{1}{2}\sum_i Z_i + \tfrac{3+\eta u}{4}\sum_i X_i \tag{D1}$$

with one jump $\sqrt{\gamma}\,(\sigma_1^- - \sigma_2^- - \sigma_3^-)$, steps of $\Delta t = 1/2$ and signs $u = \pm 1$, at $\eta = 1$ or $\eta = 8/\sqrt{\gamma}$. The jump leaves three dark states, with six slow covariances. At strong loss one step of the full master equation is a rotation of them up to corrections of order $1/\gamma$, so the mixing law without a reset (Appendix B.3) gives the first statement and the scaling limit (Appendix B.4) the second, here with explicit error bounds for the full master equation. Both end in finitely many strict inequalities, which the archive `spin_certificates.zip` in the repository checks in interval arithmetic at 256 bits (ball arithmetic of Arb, python-flint 0.9.0), and the note `three_spin_limits.pdf` writes out the derivation. An independent program builds the full finite-rate maps and agrees with the certified values to better than one part in a million. The certified rate of the exponential law lies between 61 and 133, and with the scaled amplitude at $\gamma = 2^{16}$ single inputs keep between 8% and 81% of the capacity of X_1.

Why coherence carries the memory. In the limit the mean m_0 consists of populations of the energy levels of the Hamiltonian restricted to the dark states, and a new input turns a population into a coherence between two levels: V_1 lies in the coherences. Coherence shapes the performance of spin reservoirs [52]. Dephasing in this energy basis after every step removes the coherences, and the stored signal falls to order ϵ^2, that is, as $1/\gamma^2$. Numerically the variance of X_1 stays between 10^{-4} and 2×10^{-3} from $\gamma = 2^8$ to 2^{18} when the coherences are kept and falls as γ^{-2}, by six orders of magnitude, when they are erased.

E Numerical details and supporting checks

This appendix backs the main text, section by section.

E.1 Memory lifetime and dark states (Sec. 3.1)

Figures smooth recall curves for display only, by averaging each delay with its neighbours within a fifth of the delay, and no sum uses the smoothed values. The maps of Fig. 2a,b are computed at $\gamma = 0.1$, 0.25 and 2^{-1} to 2^{6} and interpolated along the delay. Figure 2c stops the equal-phase channel and the two collective channels at $\gamma = 64$, where their recall at a fixed amplitude starts to reach past the 300 delays we sum. Summed over all delays with exact moments (Appendix A.4), their capacity keeps rising up to $\gamma = 256$.

Both tests of the dark states act on the 100 dark coordinates. For the dark-state model we write $\mathcal{P} = RR^\flat$ with $R^\flat R = 1$, where R embeds the dark coordinates α, evolve α under Eq. (A2) and record the same 45 features of the full state $(R - \mathcal{S}\mathcal{H}(u)R/\gamma)\,\alpha$. The readout is trained on them exactly as for the full reservoir, and the model has no fitted parameter (Fig. 3a). For the erasure test we replace the dark coordinates after every step by their time average, $r \mapsto r - RR^\flat(r - \bar{r})$. The control erases the dark subspace rotated by a random product of single-spin unitaries. Recall is measured at delays 1 to 100 with 800 training, 200 validation and 800 test inputs, at $\gamma = 16$ (Fig. 3b).

The decay out of the dark states falls exactly as $1/\gamma$: for the equal-phase channel $\gamma\bar{\kappa}$ approaches $\kappa_c = 2.945$ from $\gamma = 512$ on, so t_f grows in proportion to γ, while it levels off near 56 steps with the opposite-phase channel and near 110 with pairwise decay. The second-order calculation of Eq. (A2) gives the same κ_c at five spins and 1.4, 2.1 and 3.4 at three, four and six spins: at every size we computed, the decay falls as $1/\gamma$, with a prefactor that depends on the size.

Figure 6 repeats the curves of Fig. 2c at four and six spins, with 30 and 63 features, on eight and four reservoirs. The opposite-phase channel reverses the sign of the last $\lfloor N/2 \rfloor$ spins. With the equal-phase channel the capacity grows with the loss at both sizes, from 11.5 to 18.4 units between $\gamma = 1$ and 64 at four spins and from 15 to 51 at six, and a scaled amplitude raises the best capacity of the other two channels. At three spins, which have only three dark states, the capacity of the equal-phase channel does not grow.

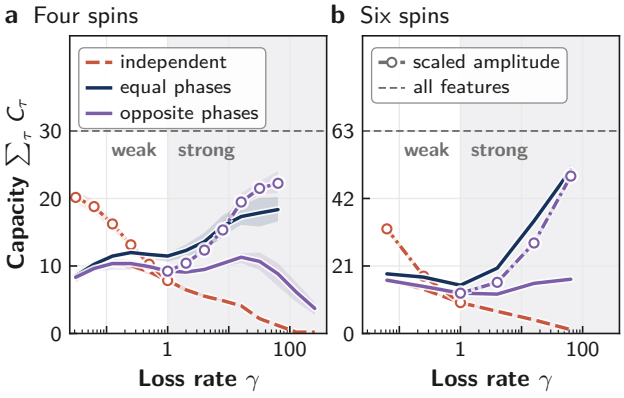


Figure 6: **Other reservoir sizes.** Capacity against the loss rate as in Fig. 2c, for four and six spins, with the same shading. A line stops where its recall reaches past the 300 delays summed.

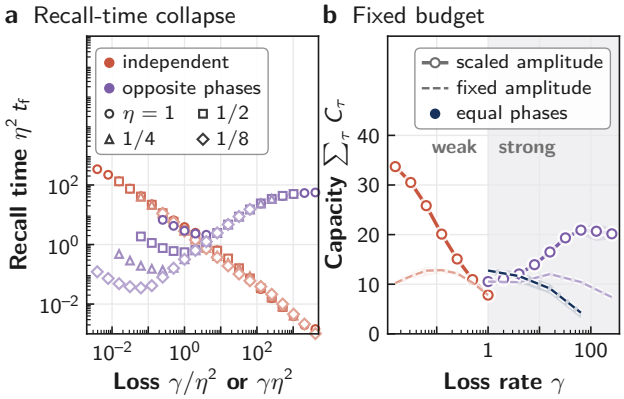


Figure 7: **Two supporting results.** **a** The recall time t_f, rescaled as the sum of decay and mixing predicts: the four amplitudes fall on one curve per channel. **b** Capacity recovered at a budget of 10^9 reservoir steps per output step, with every repeated step counted. Shading as in Fig. 2c.

E.2 Recall time and phases (Secs. 3.2 and 3.3)

The recall time t_f shows the balance of decay and mixing. Its rate adds the decay of the stored signal and a mixing rate that grows as η^2 (Appendix A.3). The decay grows as γ with independent decay and falls as $1/\gamma$ with the opposite-phase channel, so $\eta^2 t_f$ depends on loss and input only through γ/η^2 and through $\gamma\eta^2$, the $1/\nu$ and ν of Appendix B.4. Both collapses hold for the four amplitudes $\eta = 1$, $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$ (Fig. 7a), for γ from 2^{-8} to 2^{6} with independent decay and from 2^{-2} to 2^{12} with the opposite-phase channel. With the slowest mode in place of the mean rate they hold to within 11% wherever the recall time exceeds four steps.

The change per input of Fig. 4b is $2\|R^\flat[E(\frac{3}{4}) -$

$E(\frac{1}{4})]r_*\|$ at the stationary state r_* of the equal-phase channel, for the input patterns $(1, \dots, 1)$ and $\sqrt{2}\cos(2\pi i/5)$ multiplying X_i. The mode-matched pattern falls as $1/\gamma$, to 0.003 at $\gamma = 128$, while the other falls from 0.40 to 0.18. The rule is sharp: a phase error δ couples dark states directly at order δ, so beyond $\gamma \sim 1/\delta$ the equal-phase channel behaves like the opposite-phase channel.

E.3 Inputs mixed and capacities (Secs. 3.2 and 3.3)

The distribution of the number of inputs mixed, Eq. (A3), uses six-point Gauss–Legendre quadrature for the expectations over the input and solves the equations $X - \bar{A}X\bar{A}^\mathsf{T} = C$ in the eigenbasis of $\bar{A}$. The shares of the 45 features are summed until the rest falls below 10^{-8}, or until successive orders shrink at a settled ratio, whose geometric tail we add. The linear recall follows from the same moments, $C_\tau = c_\tau^\mathsf{T}\Sigma_y^{-1}c_\tau/\mathrm{Var}\,u$ with c_τ the covariance of the features with $u_{t-\tau}$ and Σ_y their covariance, and reproduces the simulated capacities to a median of 2%. Three steps of inverse iteration on $\mathcal{T}$, from the input's covariance V and from the identity, give t_mem, Q and ℓ. Loss rates run in powers of four over the ranges of Fig. 4c, and with the scaled amplitude to 16384 and 1/4096, on four reservoirs, two at the longest memory lifetimes. At a fixed amplitude the mean order of the opposite-phase channel rises from 1.8 at $\gamma = 1$ to 23 at $\gamma = 4096$, and with independent decay it stays within 2.2 of $s\,t_\mathrm{mem}$ up to $s\,t_\mathrm{mem} = 18$.

Figure 4a shows the linear recall summed over all delays, from the same moments. At a fixed amplitude it peaks at 16.0 units with the opposite-phase channel ($\gamma = 64$, $s\,t_\mathrm{mem} = 0.32$) and at 14.9 with independent decay ($\gamma = 1/16$, 0.20), and falls to 2.2 and 3.7 units at lifetimes of 750 and 430 steps, where single past inputs carry 0.05 and 0.09 of the capacity. With the scaled amplitude single inputs carry 0.82 of the capacity at a lifetime of 20 steps and 0.87 to 0.90 beyond 80 steps, the recall stays between 39 and 40 units, and $s\,t_\mathrm{mem}$ between 0.007 and 0.014. Of the inputs that the scaled amplitude removes from the mean of D with the opposite-phase channel, the mixing over the lifetime accounts for 43% at a lifetime of 20 steps, for 84% at 200 steps and for 99% at 750.

The scaling limit. We build the limit of Appendix B.4 from the generator of each reservoir: O_0, a_1 and a_2 are the orders zero to two of the step in the input field, and Y and c_0 its first order in the leak. Independent decay has 992 slow covariances and the opposite-phase channel 90, for the ten dark states under Eq. (A2), and at $\nu = 1$ they decay by at least $0.5\,\epsilon$ and $1.4\,\epsilon$ per step. With the opposite-phase channel 33 directions of the 45 features respond at leading order, 11 through the admixture of decaying operators of order $1/\gamma$, which the readout of the limit includes and which raises the share of single inputs by up to 0.02, and one at order $1/\gamma^2$, which the recursion no longer resolves from $\gamma = 1024$ on. The share of single inputs is here the share of $D = 1$, which includes the current input.

For the effective generator of Eq. (A2) the recursion converges onto the limit: at $\nu = 1$ the share of single inputs agrees to four digits from $\gamma = 2^{14}$ on. For the full master equation at $\gamma = 1/4096$ and 4096, on two reservoirs and for $\nu = 1/16$ to 64, it differs from the limit by at most 0.002 with independent decay and 0.009 with the opposite-phase channel, the mean of D by 0.003 and 0.016, and the signal, the variance that single inputs leave in the 45 features, by 0.4%. At $\nu = 1$ the limit gives single inputs 0.86 to 0.93 of the capacity over four reservoirs with independent decay and 0.90 to 0.94 with the opposite-phase channel. The signal is largest at $\nu = 22$ to 46 and at 68 to 215, where single inputs carry 0.38 to 0.55 of the capacity and $\varrho/(1-\varrho)$ lies between 0.8 and 4.8.

Table 3 collects the capacities of simulated runs on eight reservoirs. Their readout is trained and tested on a finite number of inputs, and their sum stops at 300 delays. On the four reservoirs that both use, with the scaled amplitude at $\gamma = 1/64$ and 256, the simulated runs give 35.0 and 37.1 units, and the exact moments give 36.8 and 39.4 units within the first 300 delays and 36.9 and 39.9 over all delays. All capacities of a readout add up to its rank, 45 here [29], so raising the linear memory capacity from 15 to 40 units leaves at most 5 instead of 30 for products of inputs.

E.4 Best input amplitude (Secs. 3.3 and 3.4)

We mapped the linear memory capacity over loss rates and amplitudes: the first four reservoirs, independent decay at $\gamma = 2^{-8}$ to 2^{-2} and at 1, the opposite-phase channel at $\gamma = 1$, 4 and 2^3 to

Table 3: Linear memory capacity $\sum_\tau C_\tau$ of the five-spin reservoir in simulated runs, in units (maximum 45) and summed over the first 300 delays: with exact expectation values, at a budget of 10^9 reservoir steps per output step, and with n shots per setting and step with the rewinding protocol (Sec. 3.4). Mean and standard deviation over eight reservoirs. Exact moments over all delays, as quoted in Secs. 3.2 and 3.3, are higher by up to 4 units (Appendix E.3).

	independent decay	opposite phases	equal phases
fixed amplitude[a]	14.5 ± 1.0	14.7 ± 1.1	30.6 ± 1.4
scaled amplitude[b]	34.9 ± 2.0	35.7 ± 3.0	–[c]
budget 10^9, fixed[d]	12.8 ± 0.7	12.0 ± 0.7	12.8 ± 0.7
budget 10^9, scaled[e]	33.7 ± 1.1	20.9 ± 1.1	–[c]
$\gamma = 64$, exact[f]	1.2 ± 0.0	14.7 ± 1.1	30.6 ± 1.4
$\gamma = 64$, $n = 10^4$[f]	0.2 ± 0.0	8.2 ± 0.8	0.2 ± 0.0
$\gamma = 64$, $n = 10^6$[f]	0.2 ± 0.0	9.9 ± 0.7	2.3 ± 1.0

[a] At the best loss rates, $\gamma = 0.1$, 64 and 64. [b] At $\gamma = 1/64$ and 256. [c] The uniform input is mode-matched to this channel. [d] At $\gamma = 1/8$, 16 and 1, the best loss rates on the grid. [e] At $\gamma = 1/64$ and 64. [f] At a fixed amplitude.

2^8, and fifteen amplitudes from $\eta = 2^{-7}$ to 1, with exact features or 10^6 to 10^8 shots per setting and step from the state with the full history. With exact expectation values the capacity rises as the input weakens, up to 43 of the 45 units. With 10^6 to 10^8 shots an optimum exists. Its amplitude falls with the relaxation time, here the recall time at $\eta = 2^{-8}$, with fitted exponents between -0.40 and -0.46 in all six fits over the loss rates whose relaxation time is at least four steps. Without the loss rates whose recall reaches past the 300 delays summed, the exponents are -0.48 to -0.56 with independent decay and -0.38 to -0.39 with the opposite-phase channel. The optimum is flat: one amplitude for all loss rates recovers on average 91% to 97% of the capacity at the optimum tracked at each loss rate, so the input need not be tuned to the loss rate.

E.5 Measurement (Sec. 3.4)

Weak signals cost measurements: with independent decay at $\gamma = 64$ the previous input moves the state by about 10^{-9}, which gives perfect recall of that input with exact expectation values and none with 10^8 shots per setting and step. With the opposite-phase channel, the scaled amplitude lowers the signal from 8×10^{-5} to 5×10^{-6}, and restarting each run from $|00000\rangle$, as the rewinding protocol does, raises it to 4×10^{-5}, against 1.2×10^{-4} with independent decay and the scaled amplitude at $\gamma = 1/32$.

Table 4: The eight decay channels at a relaxation time of 32 input steps and $\eta = 0.35$: loss rate, linear response of the stationary state, and linear memory capacity with exact expectation values and with 10^6 shots per measurement setting and step. Mean and standard deviation over eight reservoirs.

channel	loss rate γ	response	capacity exact	capacity at 10^6
independent	0.092	1.3	23.7 ± 1.1	17.4 ± 2.3
exchange	0.064	0.30	25.1 ± 0.8	10.1 ± 0.9
gain/loss	0.079	0.46	24.4 ± 0.8	12.5 ± 0.9
pairwise	168	1.5	18.2 ± 2.0	12.4 ± 1.2
opposite phases	101	0.68	25.3 ± 2.6	15.0 ± 1.5
two channels	73	0.011	26.6 ± 2.9	0.3 ± 0.1
equal phases	47	0.018	35.3 ± 0.8	0.3 ± 0.1
dephasing	143	0	0.0	0.2[a]

[a] A readout without any recall reads about 0.2 units, the floor of the sum over 300 delays.

The rewinding window τ_{90} grows with the memory lifetime: with the scaled amplitude of the opposite-phase channel from about 30 steps at $\gamma = 16$ to 130 at $\gamma = 256$. The thresholds of Sec. 3.4 are the crossings of the capacities at a fixed and with the scaled amplitude, from runs with the rewinding protocol at 10^3 to $10^{8.5}$ shots per setting and step. With the rewinding protocol the equal-phase channel at $\gamma = 64$ stays at the floor up to 10^4 shots and recovers 2.3 units at 10^6 and 15.5 at 3×10^8. There it passes the opposite-phase channel at a fixed amplitude, 11.0 units, and stays below it with the scaled amplitude, 26.3 units. At 10^3 and 10^4 shots the opposite-phase channel at $\gamma = 64$ recovers 5.4 and 8.2 units at a fixed amplitude, and independent decay at $\gamma = 1/32$ recovers 8.2 and 11.0. The scaled amplitude overtakes the fixed one at 2×10^4 shots with independent decay and at 10^5 with the opposite-phase channel, and independent decay then reaches 31.5 units at 3×10^8.

Figure 7b shows the capacity at a budget of 10^9 reservoir steps per output step, which is 3×10^6 to 3×10^7 shots per setting and step, and Table 3 lists its best values. With the full history, eight times more training inputs change the sampled capacity by less than 1.3 units.

For the matched relaxation time of Fig. 5b,c and Table 4 we find, for every channel and reservoir, the loss rate at which $t_{\mathrm{rel}} = 32$ steps, at weak loss when the kernel of the dissipator is the stationary state alone and at strong loss other-

wise, and run the reservoir there with $\eta = \sqrt{4/32}$. The linear response of the stationary state is the Euclidean norm of the Pauli coefficients $\mathrm{Tr}(PW)$, over all Pauli strings P, of the change W: the commutator $-i[H_{\mathrm{in}}, \rho_*]$ with $H_{\mathrm{in}} = h\sum_i X_i$ and ρ_* the stationary state, projected spectrally onto the slow modes. These are the modes of the kernel, or without one those decaying at most four times faster than the slowest. The loss rate is matched for each reservoir, so Table 4 gives its mean. Over the 56 reservoirs of the seven channels that respond, the rank correlation of the linear response is 0.96 with the recovered fraction of the capacity and 0.83 with the recovered capacity itself.

E.6 Local loss and dephasing

Adding independent decay $\sqrt{\gamma_{\mathrm{loc}}}\,\sigma_i^-$ to the equal-phase channel gives $\bar{\kappa} = \kappa_{\mathrm{c}}/\gamma + \kappa_{\mathrm{l}}\gamma_{\mathrm{loc}}$, where to first order κ_{l} is the trace of the added dissipator on the 99 slow dark modes divided by 99, $140/99 \approx 1.414$. Local dephasing $\sqrt{\gamma_\phi}\,Z_i$ adds $\kappa_\phi\gamma_\phi$ with $\kappa_\phi = 403.2/99 \approx 4.073$, and the full spectra reproduce both coefficients to within 0.2%. A recall time of at least T steps, $\kappa_{\mathrm{c}}/\gamma + \kappa_{\mathrm{l}}\gamma_{\mathrm{loc}} \le 1/(\Delta t\, T)$, then needs $\gamma/\gamma_{\mathrm{loc}} \ge 4\kappa_{\mathrm{c}}\kappa_{\mathrm{l}}\,\Delta t^2 T^2 \approx 4.2\,T^2$, or $12\,T^2$ for dephasing.

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